

Kryštof MATUŠTÍK
Machine learning student

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Master's student in **Artificial Intelligence and Machine Learning** with a strong technical foundation in Informatics.
Experienced in communicating complex concepts through **teaching university-level seminars** and evaluating low-level programming assignments.
Passionate about applying ML methods to real-world problems and eager to broaden my experiences.

SKILLS

Languages	Python (Advanced), C/C++ (Intermediate), SQL, Bash/Shell
AI & ML	PyTorch, Pandas, NumPy, Transformers, RAG, Local LLMs, Quantization
Tools	Git, Docker, UNIX, LaTeX, NeoVim, Proxmox, llama.cpp
Soft Skills	Technical Teaching, Public Speaking, Cross-cultural Communication

EXPERIENCE

Present Feb 2024	Teaching Assistant / Seminar Lead, MASARYK UNIVERSITY, Brno > Introduction to AI (PB016 Course): Led seminars for two semesters, teaching fundamental concepts of AI, search algorithms, and basic logic to undergraduate students. > Low-Level Programming (IB111 Course): Assisted in evaluating student assignments (C), providing feedback on code quality, memory management, and debugging. > Developed strong technical communication skills by breaking down complex topics for students. Teaching Python C Algorithms Public Speaking
Aug 2023 Jan 2016	Sales & Logistics Associate, VARIOUS COMPANIES, Ostrava > Wine Shop Sales: Managed customer transactions, inventory, and product advice. > Logistics Driver: Responsible for time-sensitive deliveries for a flower and food companies. > Construction & Renovation: Assisted in residential remodeling, focusing on bathroom renovations and technical on-site problem-solving. > Gained strong time management and manual skills, reliability, and ability to work in fast-paced environments. Customer Service Logistics Teamwork Manual skills Adaptability

LANGUAGES

Czech	● ● ● ● ●
English	● ● ● ● ●
German	● ● ○ ○ ○

PROGRAMMING

Python	● ● ● ● ●
C/C++	● ● ● ● ○
Git	● ● ● ● ○
UNIX	● ● ● ● ○
SQL	● ● ● ○ ○

EDUCATION

2025 - Present	Master of Science in AI and ML , Masaryk University, Brno, Czech Republic.
2025 – 2026	Erasmus exchange, University of Cagliari , Sardinia, Italy.
2022 - 2025	Bachelor of Science in Informatics , Masaryk University, Brno, Czech Republic.
2014 - 2022	Maturita state exam , Gymnázium Ostrava-Hrabůvka, Ostrava, Czech Republic.

♥ INTERESTS

- > **Local AI & Homelab:** Custom-built server with dual RTX 3060s running Proxmox; self-hosted LLM inference stack with multiple quantized models; experimenting with VRAM optimization, model routing, and open-weight architectures, running multiple VMs and LXC.
- > **Music:** Electric guitar

+ PROJECTS & VOLUNTEERING

BACHELOR THESIS: COMPUTATIONAL WORKFLOW FOR GENOME-SCALE METABOLIC MODEL RECONSTRUCTION OF NON-MODEL ORGANISMS 2025

[GitHub Link](#) [Thesis Link](#)

Collaborated with a university research group to help develop whole genome metabolic models.

- > Processed and analyzed complex biological datasets to simulate metabolic networks.
- > Wrote clean, reproducible code to validate model predictions.
- > **Tech Stack:** Python, Data Analysis, Biological Modeling.

Research Python Data Science Bioinformatics

GPT-2 LANGUAGE MODEL - FROM-SCRATCH IMPLEMENTATION 2026

[GitHub](#)

Built and trained a complete GPT-2 transformer (83M parameters) entirely from scratch in PyTorch.

- > Implemented all core components: multi-head masked self-attention, pre-LayerNorm transformer blocks, feed-forward layers with GELU activation, token and positional embeddings.
- > Pre-trained on 3.2 billion tokens (OpenWebText subset) reaching a stable cross-entropy loss of 3.75; fine-tuned for haiku style mimicry.
- > Mixed-precision (bfloat16) training loop with gradient clipping, automated loss logging, and multi-GPU support via `nn.DataParallel`.

PyTorch Transformers Attention-mechanism GPU Optimization

LINGUADAILY — AI-POWERED LANGUAGE LEARNING DAEMON 2026

[GitHub](#)

Language-learning system delivering daily lessons via Telegram.

- > Orchestrated multiple local LLMs on constrained 24GB VRAM: Gemma 4 MoE 26B Q4_K_M for translation and tutor chat, OmniVoice for TTS audio generation.
- > Built tutor chat with lesson context, flashcard quizzes, and a companion web UI for profile management, model selection, and system monitoring.
- > Implemented a RAG pipeline using grammar reference pdfs for grounded context in tutoring chat, eliminating reliance on model knowledge and limiting hallucinations.

Python RAG Local LLMs Telegram Bot API TTS Web UI

PERSONAL PROGRAMMING PROJECTS 2022 - PRESENT

[GitHub: krstff](#)

Applied advanced algorithmic concepts to build solving and predictive tools.

- > **Guitar Note Predictor (Python):** Built a Markov Chain model using Python to analyze real-time audio input and predict subsequent notes based on MIDI training data.
- > **Wordle & Spelling Bee Solver (C):** Implemented a high-performance solver in C that utilizes constraint checking and dictionary filtering to find optimal solutions efficiently.

Python C Markov Models Optimization

INTERNATIONAL VOLUNTEER PROGRAM SUMMER 2025

Italy

Volunteered for one month in Italy, contributed to a local olive farm.

- > Adapted quickly to a new cultural environment and foreign language barriers.

Volunteering Soft Skills Adaptability